

CLINICAL INTELLIGENCE TREND REPORT

YELLOW: Closer Observation Suggested

Patient ID: TMillerPeriod: July 31 to August 30, 2024Report Date: August 31, 2024Coverage: 510/744h (68%)

Decision-support summary derived from longitudinal vital sign trends; interpret in clinical context.

Clinical Alerting Thresholds

Very Low HR < 40 bpm

Very High HR > 110 bpm

Low HR < 45 bpm

Elevated Breathing > 24 brpm

Elevated HR > 95 bpm

High Breathing > 30 brpm

High HR > 100 bpm

Very High Breathing > 40 brpm

Last 24 Hours



| Vital Stat | Avg     | Min     | Max     |
|------------|---------|---------|---------|
| Heart Rate | 49 bpm  | 40 bpm  | 66 bpm  |
| Breathing  | 18 brpm | 11 brpm | 44 brpm |

Limited data in the final 24h (15/24h); values reflect available readings.

Last 24 hours: one low heart rate episode overnight, otherwise within normal range.

Episodic Events (last 24h)

| Time Span    | Duration | Episodes | Condition              | Avg | Min | Max | Comment                                                                         |
|--------------|----------|----------|------------------------|-----|-----|-----|---------------------------------------------------------------------------------|
| 2 AM to 3 AM | 1h       | 1        | Elevated Breathing     | 18  | 11  | 44  | Check SpO2 and lung sounds; assess for fluid overload or early infection        |
| 2 AM to 3 AM | 1h       | 1        | Low Heart Rate (brief) | 41  | 40  | 43  | Check pulse and BP; review beta blockers and AV node agents; ECG if symptomatic |

Patient clinical status over 31 days from Jul 31 to Aug 30

HRBreathing



42 episodic events Detected, spanning 3 days total. Conditions: Elevated Breathing, High Breathing, Low Heart Rate. Priority grade: Medium

| Episodic Events | Total Hours | Highest Burden | Episodes/day | Coverage |
|-----------------|-------------|----------------|--------------|----------|
| 42              | 78h         | Low Heart Rate | 1            | 68%      |

Major Findings: Sustained low HR (avg 43, min 40)

| Date   | Time Span      | Duration | Episodes | Condition          | Avg | Min | Max | Comment                                                                         |
|--------|----------------|----------|----------|--------------------|-----|-----|-----|---------------------------------------------------------------------------------|
| Aug 15 | 2 PM to 3 PM   | 2h       | 2        | High Breathing     | 32  | 16  | 45  | Check SpO2, work of breathing, fever; consider pulmonary cause or CHF           |
| Aug 22 | 1 PM to 2 PM   | 2h       | 2        | High Breathing     | 35  | 12  | 55  | Check SpO2, work of breathing, fever; consider pulmonary cause or CHF           |
| Aug 09 | 11 AM to 12 PM | 24h      | 10       | Low Heart Rate     | 43  | 40  | 48  | Check pulse and BP; review beta blockers and AV node agents; ECG if symptomatic |
| Aug 24 | 12 PM to 2 PM  | 28h      | 12       | Low Heart Rate     | 43  | 40  | 48  | Check pulse and BP; review beta blockers and AV node agents; ECG if symptomatic |
| Aug 08 | 10 AM to 11 AM | 6h       | 6        | Elevated Breathing | 26  | 12  | 53  | Check SpO2 and lung sounds; assess for fluid overload or early infection        |
| Aug 20 | 9 PM to 10 PM  | 5h       | 5        | Elevated Breathing | 25  | 11  | 52  | Check SpO2 and lung sounds; assess for fluid overload or early infection        |

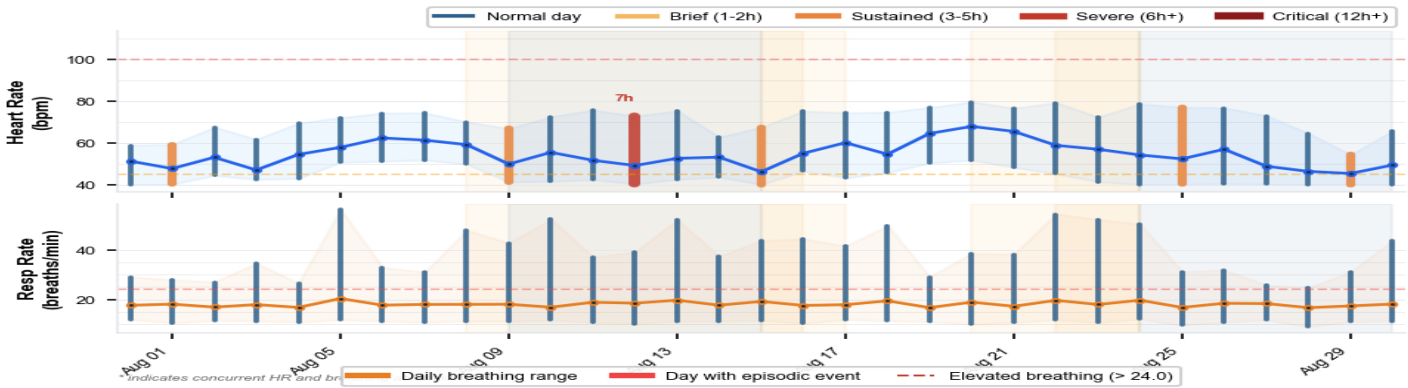
The P5 to P95 heart rate spread was 28 bpm (43 to 71), indicating significant cardiac variability.

Suggested Clinical Review Actions

- Low Heart Rate (avg 43 bpm, min 40 bpm): Check pulse and BP; review beta blockers and AV node agents; ECG if symptomatic
- Elevated Breathing (avg 26 brpm, peak 53 brpm): Check SpO2 and lung sounds; assess for fluid overload or early infection
- High Breathing (avg 32 brpm, peak 45 brpm): Check SpO2, work of breathing, fever; consider pulmonary cause or CHF
- Overall trajectory shows 6 unstable phases with 28 bpm variability. Consider increasing monitoring frequency and lowering escalation threshold.

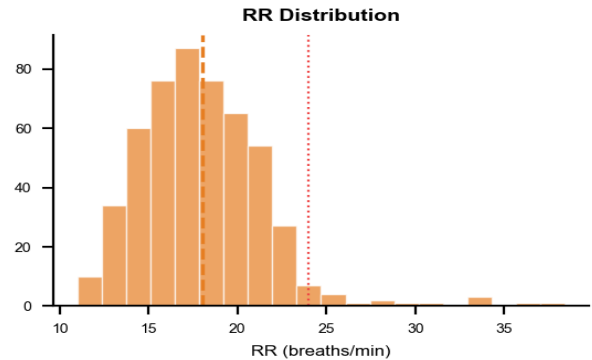
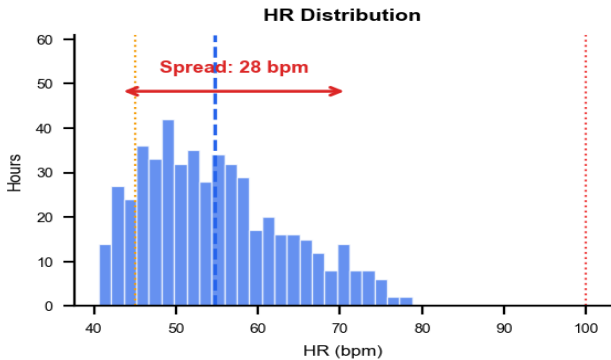
Clinical Pattern Observations:

- 1. **Sustained finding:** 7h continuous Low Heart Rate on Aug 12.
- 2. **High variability:** HR spread 28 bpm (42 to 71).
- 3. **Clustered pattern:** Episodic events on 6 of 31 days (19%).

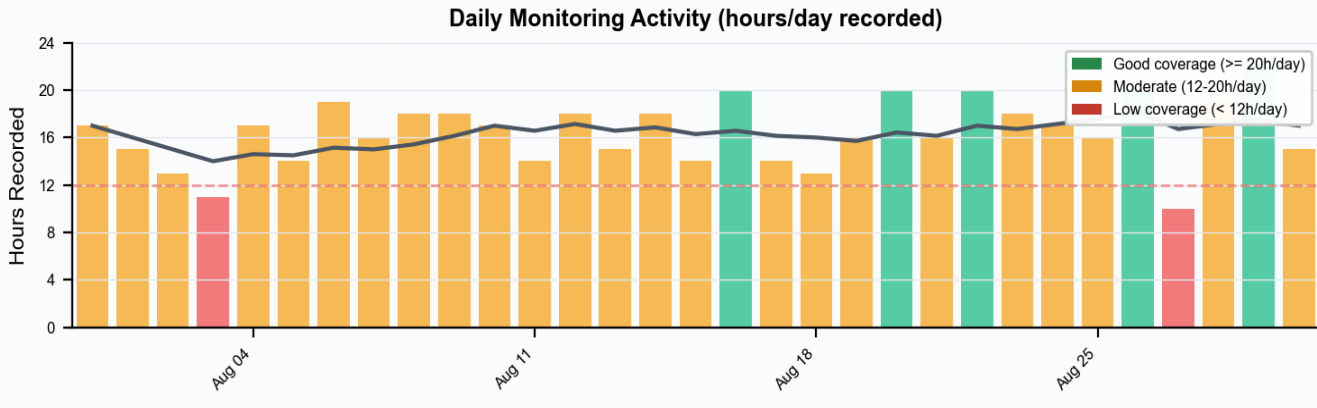


Red bars indicate days with detected episodic events.

Vital Sign Distribution



Monitoring Activity



Bars indicate monitoring hours/day. Red threshold line indicates clinical monitoring baseline.

■ HR episode ■ RR episode ■ Recorded, no episode ■ No data

Episodic event: threshold exceeded continuously for  $\geq 1$  hour. Episodes separated by  $\leq 1$  hour of normal vitals are counted as the same event. Brief threshold crossings that do not sustain are not flagged.